V4.2 Toy Model 4E
Relational Stress-Energy Tensor Candidate

Rev A - Exploratory Gravity / Metric Bridge Report

Date: 2026.05.26 | *Continuation after Toy Model 4D*

Status: Toy Model 4E builds and tests candidate stress-energy-like tensors from relational density, differential structure, pressure/tension, and shear. It does not derive physical stress-energy, Newton's constant, Einstein Field Equations, dark matter, dark energy, or physical 3T.

1. Purpose

Toy Model 4D showed that raw scalar density u is not enough to balance the Einstein-tensor candidate. Toy Model 4E upgrades the source side from a scalar bump to a rank-2 tensor candidate.

The narrow test is whether a tensorial source built from relational density, gradients, curvature of the density profile, pressure/tension, and shear can match the full G_mu_nu structure better than scalar or isotropic models.

2. Critical correction before interpretation

A good numerical fit is not a physical derivation. A flexible tensor basis can approximate G_mu_nu, but that does not prove the physical stress-energy tensor has been derived. The real target is to identify the missing structural requirements: density alone is insufficient; flow, pressure, tension, shear, and conservation must be included.

3. Tensor component meaning

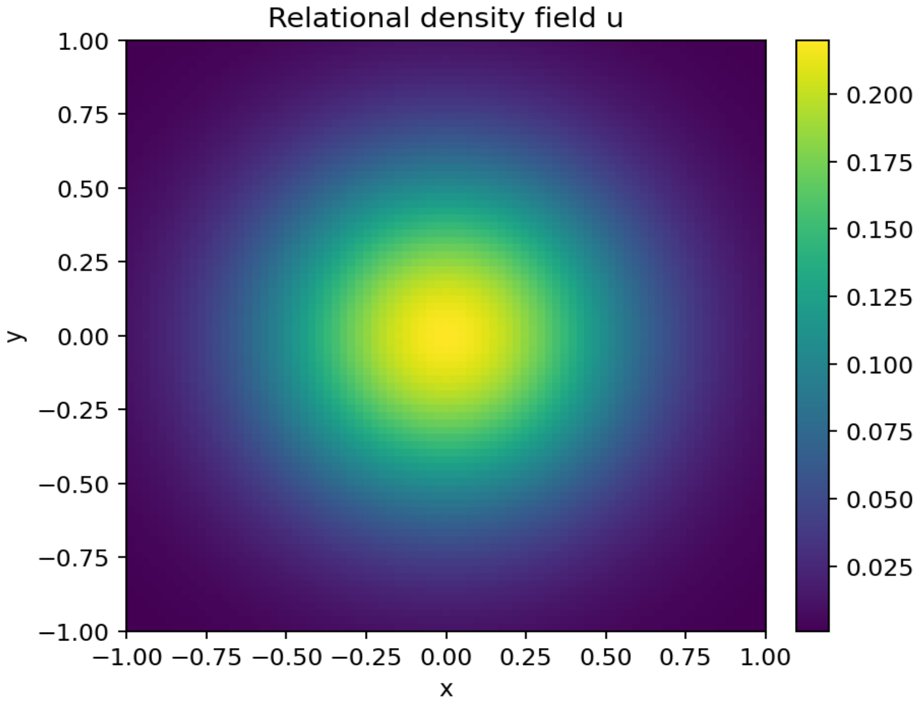
```
T_mu_nu candidate components in 1+2D:

T_tt  = energy-density-like relational concentration
T_tx  = x-flow / momentum-density-like term
T_ty  = y-flow / momentum-density-like term
T_xx  = x-direction pressure / tension
T_yy  = y-direction pressure / tension
T_xy  = shear / anisotropic relational stress
```

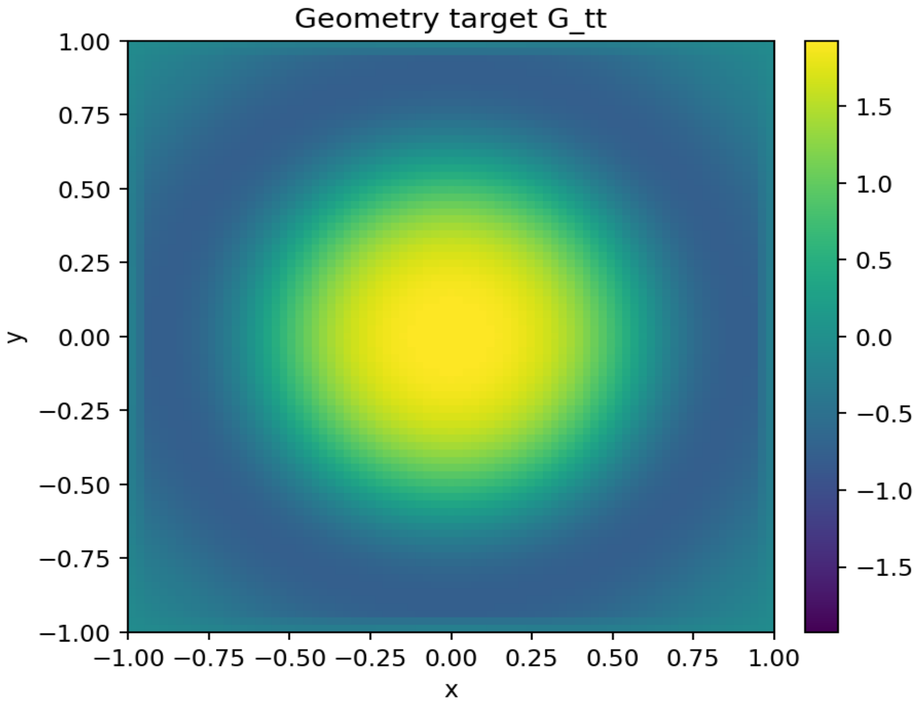
In this static symmetric test, T_tx and T_ty are set to zero because the metric and source are time-independent. A moving or rotating relational source would need nonzero flow components.

4. Source variables used

```
u              = raw relational density
grad u         = first spatial derivative / directional change
|grad u|^2     = gradient energy-like feature
laplacian(u)   = second spatial derivative / profile bending
Hessian(u)     = directional second-derivative tensor
u^2            = nonlinear density contribution
```

These are not yet physical stress-energy. They are candidate relational features from which a stress-energy-like tensor may be constructed.





5. Candidate models tested

Candidate	Definition	Purpose
Dust scalar	Only T_tt = k u; all pressure/shear/flow zero.	Tests whether raw density alone can source geometry.
Isotropic tensor	T_tt and T_xx=T_yy fitted from scalar differential features.	Tests density plus isotropic pressure.
Relational tensor	T_tt, T_xx, T_yy, T_xy fitted from density, gradients, Hessian, and nonlinear terms.	Tests density + pressure/tension + shear.
Tautological	T_mu_nu = G_mu_nu / kappa.	Diagnostic only; proves nothing physically.

6. Full tensor balance test

The main test is the full component relation:

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G_mu_nu ~= kappa T_mu_nu
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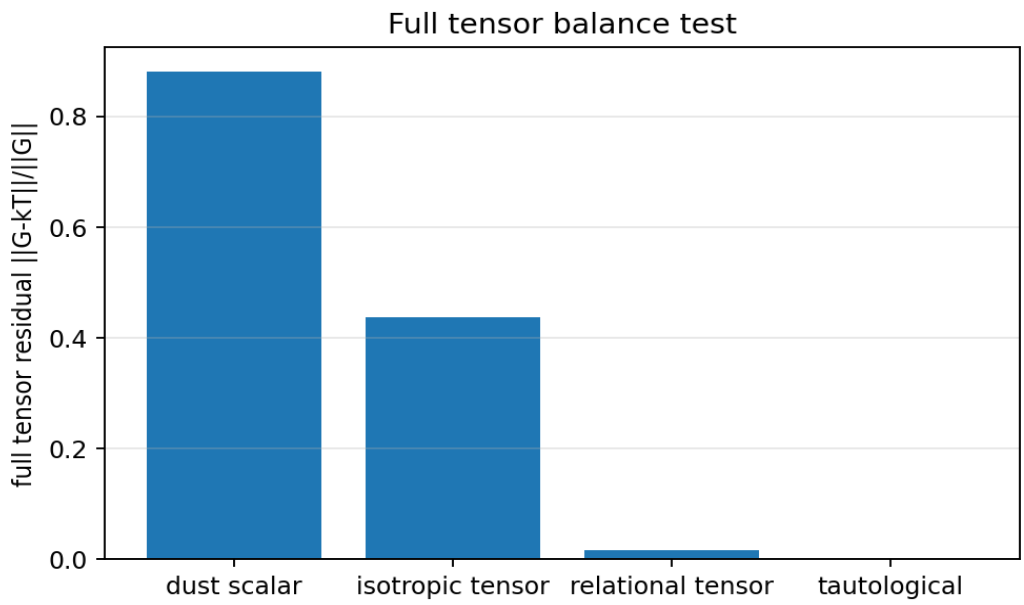
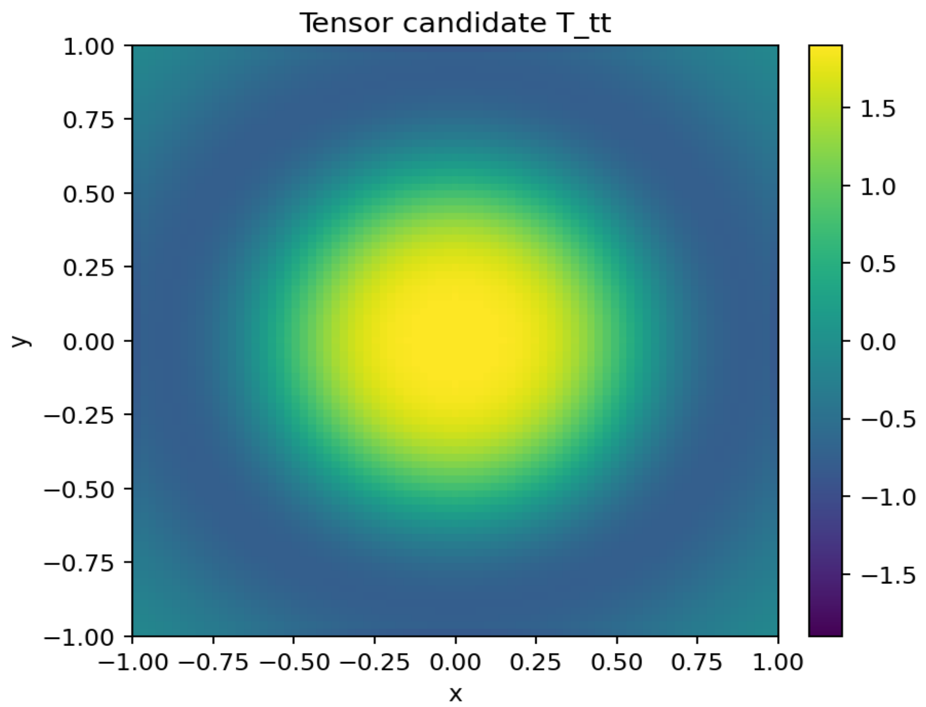
Here kappa is only a fitted toy coupling constant. It is not Newton's gravitational constant and not the physical GR coupling.

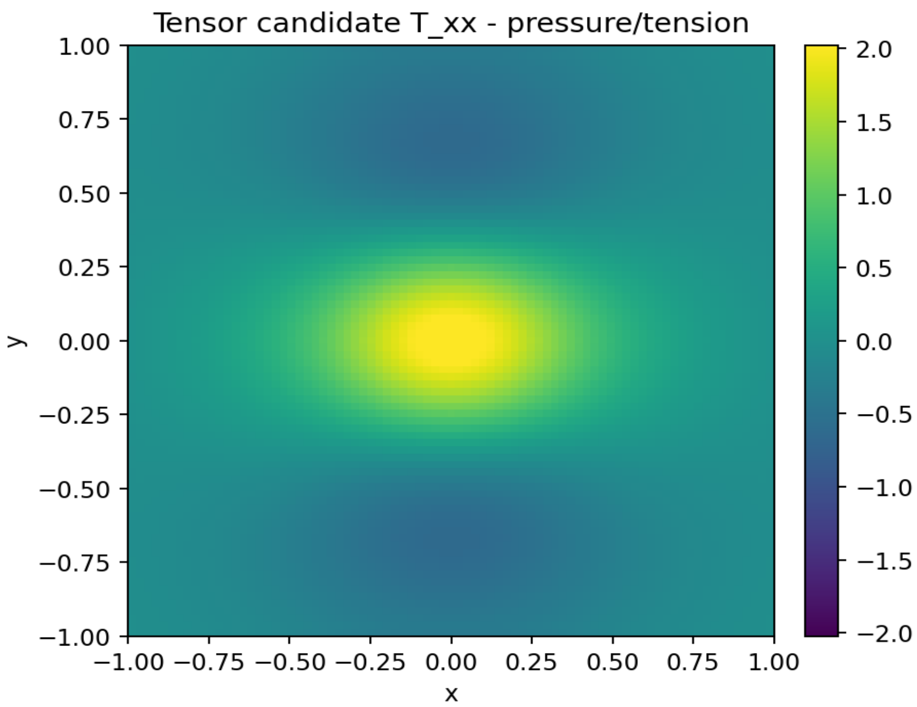


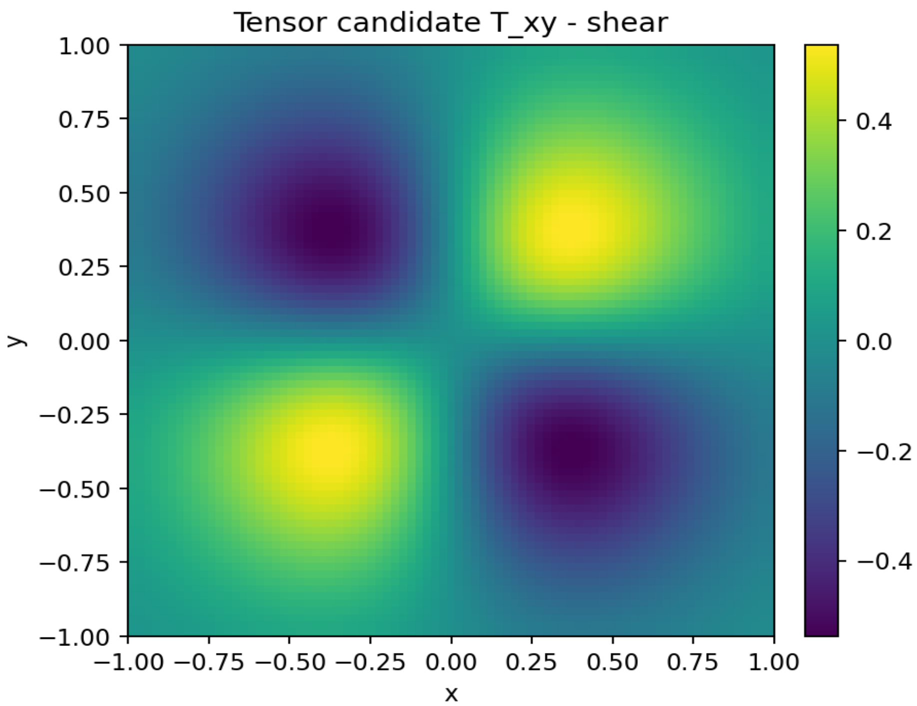
Candidate	Fitted kappa	Full tensor residual	Correlation	Max div(T)	Mean div(T)
Dust scalar	1.000000e+00	8.809001e-01	0.466452	2.774774e-01	7.894831e-02
Isotropic tensor	1.000000e+00	4.365096e-01	0.898695	3.071132e+00	1.114622e+00
Relational tensor	1.000000e+00	1.657744e-02	0.999861	1.234161e-01	7.201543e-02
Tautological	1.000000e+00	0.000000e+00	1.000000	1.299728e-02	2.481676e-03

7. Relational tensor component maps

The relational tensor candidate builds separate density, pressure/tension, and shear-like components.



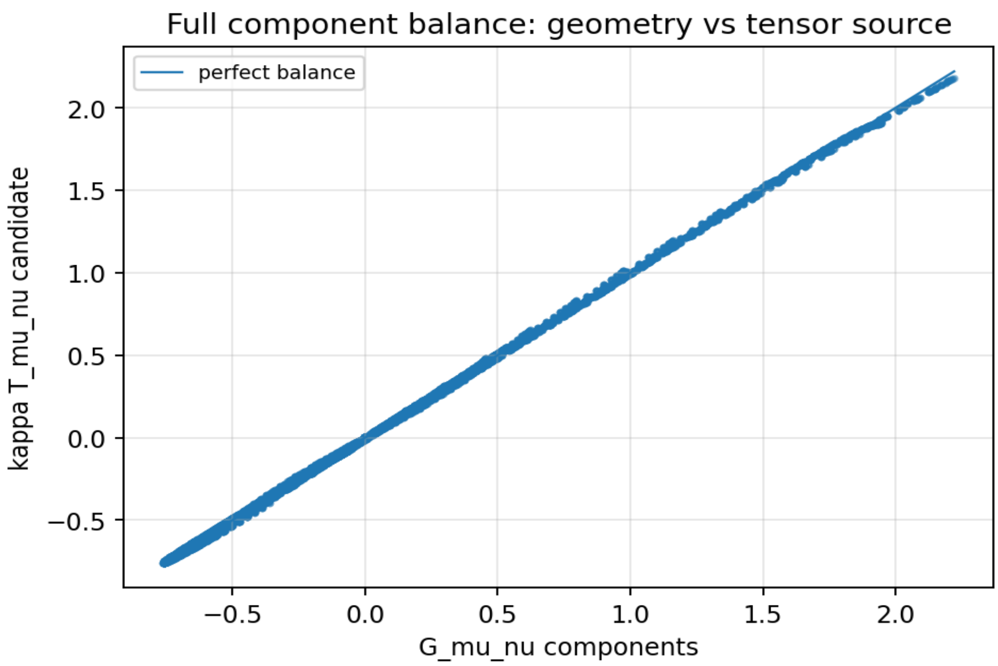




8. Component-by-component check

The full tensor candidate was checked by component. The static source correctly has no flow components in this simplified case.

Component	Residual	Correlation	Max |G|	Max |T|
tt	1.413074e-02	0.999899	1.945753e+00	1.904254e+00
tx	0.000000e+00	1.000000	0.000000e+00	0.000000e+00
ty	0.000000e+00	1.000000	0.000000e+00	0.000000e+00
xx	1.595057e-02	0.999867	2.220410e+00	2.181946e+00
xy	2.676244e-02	0.999642	5.340168e-01	5.462945e-01
yy	1.595057e-02	0.999867	2.220410e+00	2.181946e+00

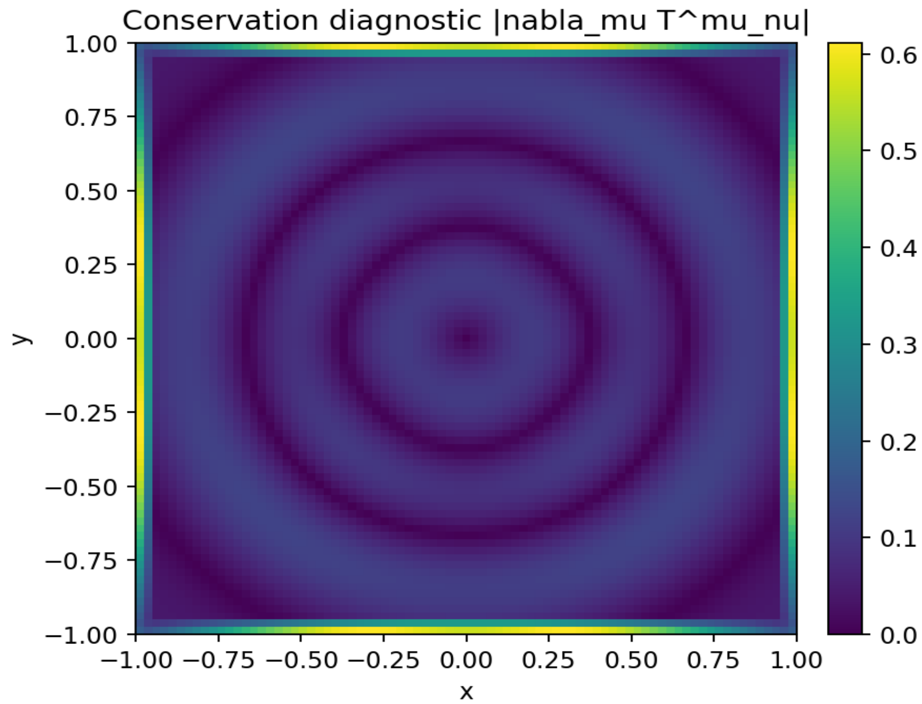


9. Conservation check

The conservation target is:

```
nabla_mu T^mu_nu ~= 0
```

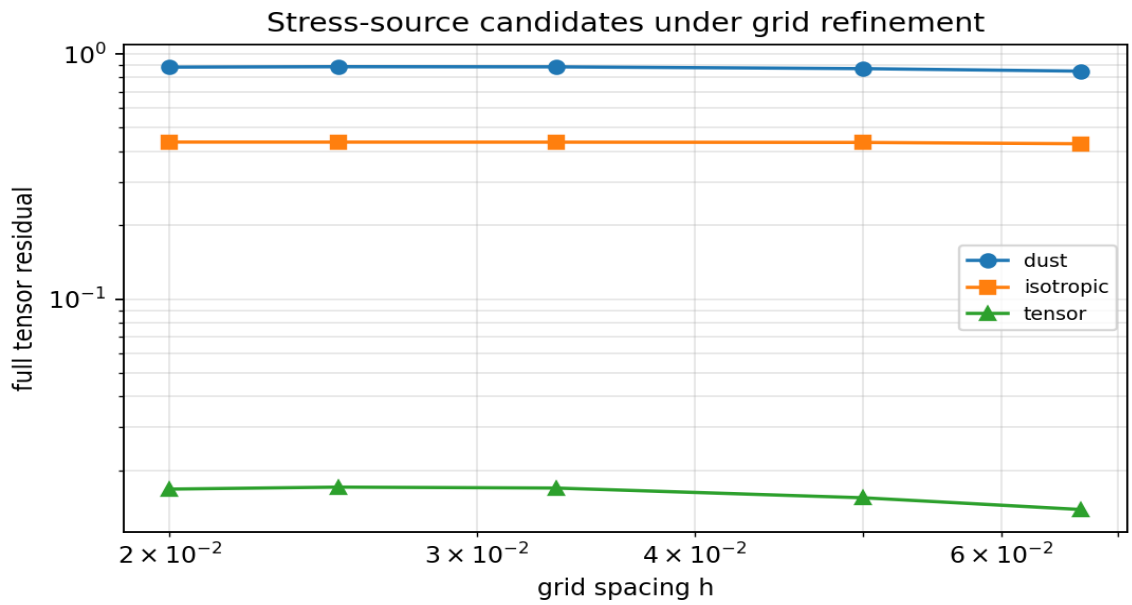
Meaning: the source tensor must be covariantly conserved if it is to balance an Einstein-tensor-like geometry. The tensor candidate has a small but nonzero finite-difference divergence. The tautological model follows G and therefore inherits the geometry-side Bianchi behavior, but it is not an independent source model.



Conservation diagnostic	Value
Max |nabla_mu G^mu_nu|	1.299728e-02
Mean |nabla_mu G^mu_nu|	2.481676e-03
Relational tensor max |nabla_mu T^mu_nu|	1.234161e-01
Relational tensor mean |nabla_mu T^mu_nu|	7.201543e-02

10. Grid-refinement behavior

The candidate models were re-fitted across multiple grid densities. The tensorial candidate remains much better than the scalar and isotropic candidates.



N	h	dust res.	isotropic res.	tensor res.	max divT	Lorentz sig.
31	0.06667	8.497594e-01	4.296355e-01	1.385039e-02	1.540494e-01	529/529
41	0.05000	8.701936e-01	4.350163e-01	1.546374e-02	1.313386e-01	1089/1089
61	0.03333	8.849279e-01	4.362265e-01	1.694247e-02	1.162700e-01	2809/2809
81	0.02500	8.860523e-01	4.363784e-01	1.709547e-02	1.161117e-01	5041/5041
101	0.02000	8.826826e-01	4.366014e-01	1.677396e-02	1.215580e-01	7569/7569

11. Physical sanity notes

Diagnostic	Value
Effective rho min	-2.622299e-01
Effective rho max	6.589114e-01
T_xx min	-6.317316e-01
T_xx max	2.181946e+00
T_yy min	-6.317316e-01
T_yy max	2.181946e+00
Max |T_xy| shear	5.462945e-01
Lorentzian signatures	4761/4761

These are sanity diagnostics only. They are not physical energy-condition proofs.

12. Interpretation inside the V4.2 framework

Toy Model 4E supports this limited ladder:

```
scalar relational density u
      -> insufficient scalar source

relational density + derivatives + tension/shear structure
      -> rank-2 tensor source candidate T_mu_nu
      -> much better component balance with G_mu_nu
      -> approximate conservation check
      -> need for true relational stress-energy law
```

It does not yet support this stronger ladder:

```
network relations
      -> physical T_mu_nu
      -> Einstein Field Equations
      -> Newton's G
      -> real gravity / dark sector solution
```

13. What Toy Model 4E achieves

• It upgrades the source side from a scalar to a rank-2 tensor candidate.

• It explicitly tests density, isotropic pressure, tension, shear, and differential features.

- It shows scalar dust performs poorly compared with tensorial relational stress.
- It shows a tensorial differential source can closely match the full G_mu_nu candidate in this controlled model.
- It computes a conservation diagnostic for the source tensor.
- It confirms that the next missing object is not merely a tensor fit, but a principled relational stress-energy law.

14. What Toy Model 4E does not achieve

- It does not derive the physical stress-energy tensor.
- It does not derive Einstein Field Equations.
- It does not derive Newton's gravitational constant or the GR coupling constant.
- It does not derive matter fields or equations of state.
- It does not prove that relational-network stress physically causes curvature.
- It does not solve gravity, dark matter, or dark energy.
- It does not prove physical 3T.

15. Close-out statement for 4E

Toy Model 4E confirms the lesson from 4D: a scalar relational-density bump is not enough. The source side must be tensorial and differential. A relational stress-energy-like tensor built from density, gradients, Hessian/tension, and shear features can approximate the full Einstein-tensor candidate much better than scalar or isotropic models in this controlled test. However, because the tensor is calibrated from a flexible feature basis, this is not a physical derivation. The correct next task is to define a principled relational stress-energy law with conservation built in, rather than merely fitting G_mu_nu.

16. Recommended next chapter

The next logical chapter is Toy Model 4F: Conservation-Built Relational Stress Law.

1. Define a relational current J^mu and stress tensor T_mu_nu from network flow and tension rules.
2. Build conservation into the source: nabla_mu T^mu_nu = 0 or a discrete equivalent.
3. Test whether the conserved source can still balance G_mu_nu component-by-component.
4. Only after a conserved source law exists should an Einstein-equation analogue be discussed.
5. Dark-sector interpretation remains premature until this source law exists.

17. Rebind prompt for continuation

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We are resuming after Toy Model 4E.

4D result:
Raw scalar density u is correlated with G_tt but fails strict proportionality.
Differential/composite sources perform better.

4E result:
The source was upgraded from scalar S to a rank-2 tensor candidate T_mu_nu.

Components:
T_tt = density-like relational concentration
T_tx, T_ty = flow / momentum-density-like terms
T_xx, T_yy = pressure/tension
T_xy = shear / anisotropic relational stress

Candidates tested:
1. Dust scalar: only T_tt = k u.
2. Isotropic tensor: T_tt and T_xx=T_yy from scalar differential features.
3. Relational tensor: density + gradients + Hessian/tension + shear features.
4. Tautological T=G/kappa: diagnostic only, not a source law.

Main result:
Scalar dust performs poorly.
Relational tensor performs much better in the controlled model.
But flexible tensor fitting is not a physical derivation.

Critical guardrail:
4E does not derive physical stress-energy, Einstein Field Equations, Newton's G,
gravity, dark matter, dark energy, or physical 3T.

Main lesson:
The next missing object is a principled conserved relational stress-energy law,
not another fitted scalar source.
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Next recommended chapter:
Toy Model 4F - Conservation-Built Relational Stress Law.